

Linux Security Monitoring and Auditing

WEEK 4 Practical Lab

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Executive Summary

The purpose of this assessment was to evaluate the effectiveness of monitoring and auditing controls within an Ubuntu Linux environment. The review focused on audit configuration, event monitoring, log analysis, security assessment, governance mapping, and security control monitoring.

Several monitoring tools were examined, including Auditd, ausearch, aureport, journalctl, and Lynis. While the required auditing tools were successfully installed and configured, some limitations were identified within the laboratory environment. Audit logs, authentication logs, and journal entries were unavailable, restricting the ability to fully validate certain monitoring activities.

A Lynis security assessment was performed to evaluate the overall security posture of the system. The assessment generated a Hardening Index score of 62 and identified several security weaknesses, including vulnerable packages, password management weaknesses, lack of malware protection, absence of file integrity monitoring, and insufficient logging controls.

Overall, the system demonstrated a moderate level of security maturity. Several improvements should be implemented to strengthen monitoring, auditing, and governance controls.

1. Scope and Authorization

Scope

The assessment was performed within the IC DFA Ubuntu Practice Hub environment and included the following activities:

- Verification of Auditd installation and configuration
- Creation and review of audit monitoring rules
- Security event generation
- Linux log analysis
- Security assessment using Lynis
- Governance and control monitoring review
- Remediation planning and retesting recommendations

Authorization

All activities were conducted within the authorized training environment provided through the IC DFA Ubuntu Practice Hub and were limited to approved educational exercises.

2. Methodology

The assessment followed a structured approach consisting of four phases.

Phase 1 – Audit Configuration and Verification

Auditd and supporting audit tools were verified to determine whether security event monitoring capabilities were available.

Commands reviewed included:

auditctl
ausearch
aureport

Custom audit rules were created for:

- Monitoring changes to /etc/passwd
- Monitoring changes to /etc/shadow
- Monitoring program execution events
- Monitoring authentication-related activity

Phase 2 – Security Event Generation

Test activities were performed to generate auditable events and validate monitoring controls.

Examples included:

nano /etc/passwd
ls /tmp

These actions were intended to trigger configured audit rules and generate audit records.

Phase 3 – Linux Log Analysis

Linux logging mechanisms were reviewed using:
journalctl

The /var/log directory and related log locations were examined to verify the availability of security logging data.

Phase 4 – Security Assessment

The Lynis security auditing tool was installed and executed to assess system security posture and hardening effectiveness.

sudo lynis audit system

3. Evidence Bundle 1 – Auditd Configuration and Events**Objective**

To verify that Linux auditing tools were installed and capable of supporting security monitoring activities.

Observations

```
ICDFA Smart Cyber Shell -- valid: green - unknown: red - suggestions: grey
student@student000317:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000317:~$
```

Figure 1: Auditd Status Verification and Package Review

```
ICDFA Smart Cyber Shell -- valid: green - unknown: red - suggestions: grey
student@student000317:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000317:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@student000317:~$
```

Figure 2 : Auditd Installation Verification

```

ICDRA Smart Cyber Shell - valid: green - unknown: red - suggestions: grey
student@student000317:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000317:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded, Run 'apt list --upgradable' to see them.
student@student000317:~$ sudo apt install auditd audispd-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000317:~$

```

Figure 3 : Environmental Limitation During Auditd Validation

```

ICDRA Smart Cyber Shell - valid: green - unknown: red - suggestions: grey
student@student000317:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000317:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded, Run 'apt list --upgradable' to see them.
student@student000317:~$ sudo apt install auditd audispd-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000317:~$ which auditctl
/usr/sbin/auditctl
student@student000317:~$

```

Figure 4: Auditctl Verification and Root Access Validation

Attempted verification of the Auditd service and package status. The environment returned a systemd limitation message during validation.

Auditd and audispd-plugins were successfully installed and available within the environment.

The auditctl binary was successfully located at:

/usr/sbin/auditctl

Root access was successfully obtained to perform administrative verification activities.
Custom audit rules were created to monitor:

- Password-file modifications
- Shadow-file modifications
- Program executions
- Authentication-related activities

```
ICDFA Smart Cyber Shell – valid: green · unknown: red · suggestions: grey
student@student000317:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000317:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@student000317:~$ sudo apt install auditd audispd-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000317:~$ which auditctl
/usr/sbin/auditctl
student@student000317:~$ sudo -i

ICDFA Ubuntu Practice Hub
Use this workspace only for authorised learning and practical exercises.

root@student000317:~# id
uid=0(root) gid=0(root) groups=0(root)
root@student000317:~# auditctl -s
You must be root to run this program.
root@student000317:~# mkdir -p /etc/audit/rules.d
root@student000317:~# nano /etc/audit/rules.d/custom.rules
root@student000317:~# cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~#
```

Figure 5: Audit Rule Configuration and Validation Limitation

Custom audit rules were configured successfully; however, the Ubuntu Practice Hub environment restricted direct validation of active audit rules and returned an "Operation not permitted" error.

Environmental Limitation

Auditd was installed successfully and root access was verified. However, the ICDFA Ubuntu Practice Hub environment restricted access to the kernel audit subsystem, preventing normal auditctl status checks. This was documented as an environmental constraint rather than a configuration failure.

```
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus. Must be root.
student@studentvuu11:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@studentvuu11:~$ sudo apt install auditd audispd-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000317:~$ which auditctl
/usr/sbin/auditctl
student@student000317:~$ sudo -i

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root@studentvuu11:~# id
uid=0(root) gid=0(root) groups=0(root)
root@student000317:~# auditctl -s
You must be root to run this program.
root@student000317:~# mkdir -p /etc/audit/rules.d
root@studentvuu11:~# nano /etc/audit/rules.d/custom.rules
root@student000317:~# cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@studentvuu11:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdfa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~#
```

Figure 6: Security Event Generation Activity

Test activities were performed to generate auditable events for validation of the configured audit-monitoring controls.

Finding F01 – Audit Log Data Unavailable Evidence

```
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@student000317:~$ sudo apt install auditd audispd-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000317:~$ which auditctl
/usr/sbin/auditctl
student@student000317:~$ sudo -i

ICDFA Ubuntu Practice Hub
Use this workspace only for authorised learning and practical exercises.

root@student000317:~# id
uid=0(root) gid=0(root) groups=0(root)
root@student000317:~# auditctl -s
You must be root to run this program.
root@student000317:~# mkdir -p /etc/audit/rules.d
root@student000317:~# nano /etc/audit/rules.d/custom.rules
root@student000317:~# cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdfa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~#
```

Figure :7 Ausearch and Aureport Verification

Verification that the ausearch and aureport utilities were installed and available for audit investigation and reporting purposes.

Execution of the following commands:

ausearch -k passwd_changes and aureport

returned:

Error opening /var/log/audit/audit.log

No such file or directory

Observation

The ausearch utility was available; however, the Ubuntu Practice Hub environment did not contain /var/log/audit/audit.log. Consequently, audit event retrieval could not be performed using standard Auditd reporting methods. This was documented as an environmental limitation rather than a failure of tool installation or configuration.

```
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@student000317:~$ sudo apt install auditd audispd-plugins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000317:~$ which auditctl
/usr/sbin/auditctl
student@student000317:~$ sudo -i

ICDFA Ubuntu Practice Hub
Use this workspace only for authorised learning and practical exercises.

root@student000317:~# id
uid=0(root) gid=0(root) groups=0(root)
root@student000317:~# auditctl -s
You must be root to run this program.
root@student000317:~# mkdir -p /etc/audit/rules.d
root@student000317:~# nano /etc/audit/rules.d/custom.rules
root@student000317:~# cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdfa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~#
```

Figure 8: Missing Audit Log Evidence

Both ausearch and aureport were unable to retrieve audit information because /var/log/audit/audit.log was unavailable within the Ubuntu Practice Hub environment.

Risk Rating

High

Control Owner

Linux Administrator

Recommendation

Ensure audit logging services generate and retain audit records to support traceability, monitoring, and forensic investigations.

4. Evidence Bundle 2 – Linux Log Analysis

Objective

To review available Linux logs and identify evidence of monitoring and security-related activities.

Observations

```
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000317:~$ which auditctl
/usr/sbin/auditctl
student@student000317:~$ sudo -i

ICDFA Ubuntu Practice Hub
Use this workspace only for authorised learning and practical exercises.

root@student000317:~# id
uid=0(root) gid=0(root) groups=0(root)
root@student000317:~# auditctl -s
You must be root to run this program.
root@student000317:~# mkdir -p /etc/audit/rules.d
root@student000317:~# nano /etc/audit/rules.d/custom.rules
root@student000317:~# cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdfa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# exit
logout
student@student000317:~$ journalctl --since "today"
No journal files were found.
-- No entries --
student@student000317:~$
```

Figure 9: journalctl No Entries

No journal entries were available for review during the assessment.

The following command was executed:
`journalctl --since "today"`

The result showed:

No journal files were found

Review of the system log directories showed:

```

root@student000317:~# cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdffa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# exit
logout
student@student000317:~$ journalctl --since "today"
No journal files were found.
-- No entries --
student@student000317:~$ ls -l /var/log
total 536
lrwxrwxrwx 1 root root          39 Aug 16 03:16 README -> ../../usr/share/doc/systemd/README.logs
-rw-r--r-- 1 root root       17342 Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root        4096 Oct  1 18:40 apt
drwxr-xr-x 2 root adm         4096 Jul 13 20:10 audit
-rw-r--r-- 1 root root       61229 Jul 30 14:19 bootstrap.log
-rw-r--r-- 1 root utmp          0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root    441344 Oct  1 18:40 dpkg.log
-rw-r--r-- 1 root root          0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4096 Aug 16 03:16 journal
-rw-r--r-- 1 root utmp          0 Jul 30 14:19 lastlog
drwx----- 2 root root         4096 Aug 16 03:16 private
drwxr-xr-x 2 root adm         4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 wtmp
student@student000317:~$

```

Figure 10: /var/log Directory Review

Review of available system log files and logging directories within the Ubuntu environment.

/var/log/audit

```

-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdffa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# exit
logout
student@student000317:~$ journalctl --since "today"
No journal files were found.
-- No entries --
student@student000317:~$ ls -l /var/log
total 536
lrwxrwxrwx 1 root root          39 Aug 16 03:16 README -> ../../usr/share/doc/systemd/README.logs
-rw-r--r-- 1 root root       17342 Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root        4096 Oct  1 18:40 apt
drwxr-xr-x 2 root adm         4096 Jul 13 20:10 audit
-rw-r--r-- 1 root root       61229 Jul 30 14:19 bootstrap.log
-rw-r--r-- 1 root utmp          0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root    441344 Oct  1 18:40 dpkg.log
-rw-r--r-- 1 root root          0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4096 Aug 16 03:16 journal
-rw-r--r-- 1 root utmp          0 Jul 30 14:19 lastlog
drwx----- 2 root root         4096 Aug 16 03:16 private
drwxr-xr-x 2 root adm         4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 wtmp
student@student000317:~$ ls -l /var/log/audit
total 0
student@student000317:~$

```

Figure 11: Audit Directory Contents

The audit directory existed but contained no audit records

contained no audit records.

Additionally:

```

-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth.failures
root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdfa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# exit
logout
student@student000317:~$ journalctl --since "today"
No journal files were found.
-- No entries --
student@student000317:~$ ls -l /var/log
total 536
lrwxrwxrwx 1 root root          39 Aug 16 03:16 README -> ../../usr/share/doc/systemd/README.logs
-rw-r--r-- 1 root root       17342 Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root        4096 Oct  1 18:40 apt
drwxr-x-- 2 root adm         4096 Jul 13 20:10 audit
-rw-r--r-- 1 root root       61229 Jul 30 14:19 bootstrap.log
-rw-rw---- 1 root utmp          0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root     441344 Oct  1 18:40 dpkg.log
-rw-r--r-- 1 root root          0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4096 Aug 16 03:16 journal
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 lastlog
drwx----- 2 root root         4096 Aug 16 03:16 private
drwxr-x-- 2 root adm         4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 wtmp
student@student000317:~$ ls -l /var/log/audit
total 0
student@student000317:~$ ls -l /var/log/auth.log
ls: cannot access '/var/log/auth.log': No such file or directory
student@student000317:~$ 

```

Figure 12: Missing Authentication Log

The auth.log file was unavailable within the Ubuntu Practice Hub environment.

```

root@student000317:~# auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdfa-zsh-1000
root@student000317:~# which ausearch
/usr/sbin/ausearch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# exit
logout
student@student000317:~$ journalctl --since "today"
No journal files were found.
-- No entries --
student@student000317:~$ ls -l /var/log
total 536
lrwxrwxrwx 1 root root          39 Aug 16 03:16 README -> ../../usr/share/doc/systemd/README.logs
-rw-r--r-- 1 root root       17342 Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root        4096 Oct  1 18:40 apt
drwxr-x-- 2 root adm         4096 Jul 13 20:10 audit
-rw-r--r-- 1 root root       61229 Jul 30 14:19 bootstrap.log
-rw-rw---- 1 root utmp          0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root     441344 Oct  1 18:40 dpkg.log
-rw-r--r-- 1 root root          0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4096 Aug 16 03:16 journal
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 lastlog
drwx----- 2 root root         4096 Aug 16 03:16 private
drwxr-x-- 2 root adm         4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 wtmp
student@student000317:~$ ls -l /var/log/audit
total 0
student@student000317:~$ ls -l /var/log/auth.log
ls: cannot access '/var/log/auth.log': No such file or directory
student@student000317:~$ ls -l /var/log/syslog
ls: cannot access '/var/log/syslog': No such file or directory
student@student000317:~$ 

```

Figure 13: Missing Syslog

The syslog file was unavailable within the Ubuntu Practice Hub environment.

```

There was an error while processing parameters
root@student000317:~# nano /etc/passwd
root@student000317:~# ls /tmp
icdfa-zsh-1000
root@student000317:~# which auserch
/usr/sbin/auserch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# auserch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# exit
logout
student@student000317:~$ journalctl --since "today"
No journal files were found.
-- No entries --
student@student000317:~$ ls -l /var/log
total 536
lrwxrwxrwx 1 root root          39 Aug 16 03:16 README -> ../../usr/share/doc/systemd/README.logs
-rw-r--r-- 1 root root       17342 Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root        4096 Oct  1 18:40 apt
drwxr-xr-x 2 root adm         4096 Jul 13 20:10 audit
-rw-r--r-- 1 root root       61229 Jul 30 14:19 bootstrap.log
-rw-rw---- 1 root utmp          0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root     441344 Oct  1 18:40 dpkg.log
-rw-r--r-- 1 root root          0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4096 Aug 16 03:16 journal
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 lastlog
drwx----- 2 root root         4096 Aug 16 03:16 private
drwxr-xr-x 2 root adm         4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 wtmp
student@student000317:~$ ls -l /var/log/audit
total 0
student@student000317:~$ ls -l /var/log/auth.log
ls: cannot access '/var/log/auth.log': No such file or directory
student@student000317:~$ ls -l /var/log/syslog
ls: cannot access '/var/log/syslog': No such file or directory
student@student000317:~$ ls -l /var/log/journal
total 0
student@student000317:~$ 

```

Figure 14: Empty Journal Directory

The journal directory existed but contained no persistent journal records.

```

icdfa-zsh-1000
root@student000317:~# which auserch
/usr/sbin/auserch
root@student000317:~# which aureport
/usr/sbin/aureport
root@student000317:~# auserch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
root@student000317:~# exit
logout
student@student000317:~$ journalctl --since "today"
No journal files were found.
-- No entries --
student@student000317:~$ ls -l /var/log
total 536
lrwxrwxrwx 1 root root          39 Aug 16 03:16 README -> ../../usr/share/doc/systemd/README.logs
-rw-r--r-- 1 root root       17342 Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root        4096 Oct  1 18:40 apt
drwxr-xr-x 2 root adm         4096 Jul 13 20:10 audit
-rw-r--r-- 1 root root       61229 Jul 30 14:19 bootstrap.log
-rw-rw---- 1 root utmp          0 Jul 30 14:19 btmp
-rw-r--r-- 1 root root     441344 Oct  1 18:40 dpkg.log
-rw-r--r-- 1 root root          0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4096 Aug 16 03:16 journal
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 lastlog
drwx----- 2 root root         4096 Aug 16 03:16 private
drwxr-xr-x 2 root adm         4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp          0 Jul 30 14:19 wtmp
student@student000317:~$ ls -l /var/log/audit
total 0
student@student000317:~$ ls -l /var/log/auth.log
ls: cannot access '/var/log/auth.log': No such file or directory
student@student000317:~$ ls -l /var/log/syslog
ls: cannot access '/var/log/syslog': No such file or directory
student@student000317:~$ ls -l /var/log/journal
total 0
student@student000317:~$ last

wtmp begins Thu Aug  6 04:28:58 2026
student@student000317:~$ 

```

Figure 15: Last Command Output

Review of recorded user-session activity using the last command.

/var/log/auth.log and /var/log/syslog

were not available.

Finding F02 – Authentication and System Logs Unavailable

Risk Significance

Authentication logs are important for investigating login attempts, privileged-user activity, and potential security incidents. The absence of these logs reduces visibility into user activities and system events.

Risk Rating

High

Control Owner

Linux Administrator

Recommendation

Enable authentication and system logging to ensure that security-relevant events can be reviewed and investigated when required.

5. Evidence Bundle 3 – Lynis Security Assessment Objective

```
drwxr-x--- 2 root adm          4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp         0 Jul 30 14:19 wtmp
student@student000317:~$ ls -l /var/log/audit
total 0
student@student000317:~$ ls -l /var/log/auth.log
ls: cannot access '/var/log/auth.log': No such file or directory
student@student000317:~$ ls -l /var/log/syslog
ls: cannot access '/var/log/syslog': No such file or directory
student@student000317:~$ ls -l /var/log/journal
total 0
student@student000317:~$ last

wtmp begins Thu Aug 6 04:28:58 2026
student@student000317:~$ sudo apt install lynis -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-110n gksu
  | kde-cli-tools | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 91 not upgraded.
Need to get 602 kB of archives.
After this operation, 3202 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 lynis all 3.0.9-1 [226 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 menu amd64 2.1.50 [377 kB]
Fetched 602 kB in 2s (293 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package lynis.
(Reading database ... 41145 files and directories currently installed.)
Preparing to unpack .../archives/lynis_3.0.9-1_all.deb ...
Unpacking lynis (3.0.9-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../archives/menu_2.1.50_amd64.deb ...
Unpacking menu (2.1.50) ...
Setting up lynis (3.0.9-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynis.timer → /usr/lib/systemd/system/lynis.timer.
Setting up menu (2.1.50) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for menu (2.1.50) ...
student@student000317:~$
```

Figure 16: Lynis Installation verification

Lynis was successfully installed and made available within the Ubuntu Practice Hub environment for security assessment activities.

```

student@student000317:~$ ls -l /var/log/audit
total 0
student@student000317:~$ ls -l /var/log/auth.log
ls: cannot access '/var/log/auth.log': No such file or directory
student@student000317:~$ ls -l /var/log/syslog
ls: cannot access '/var/log/syslog': No such file or directory
student@student000317:~$ ls -l /var/log/journal
total 0
student@student000317:~$ last

wtm begins Thu Aug 6 04:28:58 2026
student@student000317:~$ sudo apt install lynis -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n gksu
  | kde-cli-tools | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 91 not upgraded.
Need to get 602 kB of archives.
After this operation, 3202 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 lynis all 3.0.9-1 [226 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 menu amd64 2.1.50 [377 kB]
Fetched 602 kB in 2s (293 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package lynis.
(Reading database ... 41145 files and directories currently installed.)
Preparing to unpack .../archives/lynis_3.0.9-1_all.deb ...
Unpacking lynis (3.0.9-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../archives/menu_2.1.50_amd64.deb ...
Unpacking menu (2.1.50) ...
Setting up lynis (3.0.9-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynis.timer → /usr/lib/systemd/system/lynis.timer.
Setting up menu (2.1.50) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for menu (2.1.50) ...
student@student000317:~$ lynis --version
3.0.9
student@student000317:~$

```

Figure 17: Lynis Version Verification

The installed Lynis version was verified before performing the system security audit. To evaluate the overall security posture of the Ubuntu environment using the Lynis auditing framework.

```

Hardening index : 62 [##### ]
Tests performed : 232
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

=====

Exceptions found
Some exceptional events or information was found!

What to do:
You can help by providing your log file (/var/log/lynis.log).
Go to https://cisofy.com/contact/ and send your file to the e-mail address listed

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)
student@student000317:~$

```

Figure 18: Lynis System Audit Results

The Lynis system audit completed successfully and provided an overall assessment of the Linux system's security configuration and hardening posture.

Assessment Results

Security Measure	Result
Lynis Version	30.9
Hardening Index	62
Tests Performed	232
Plugins Enabled	1
Firewall	Enabled
Malware Scanner	Not Installed

The Hardening Index score of 62 indicates that the system has a reasonable security baseline but requires additional hardening to align with security best practices.

```
1.0.18-lubuntu0.24.04.1|libsource-highlight-common,3.1.9-4.3build1|libsource-highlight4t64;amd64,3.1.9-4.3build1|libspandsp2t64;amd64,0.0.6+dfsg-2.1build1|libspeedsp1;amd64,1.2.1-lub
untu3|libsqlite3-0;amd64,3.45.1-lubuntu2.7|libss2;amd64,1.47.0-2.4-exp1ubuntu4.1|libssh-4;amd64,0.10.6-2ubuntu0.4|libssh-gcrypt-4;amd64,0.10.6-2ubuntu0.4|libssh2-1t64;amd64,1.11.0-4.1
ubuntu0.24.04.3|libssl3t64;amd64,3.0.13-0ubuntu3.12|libstdc++-13-dev;amd64,13.3.0-6ubuntu2~24.04.1|libstdc++6;amd64,14.2.0-4ubuntu2~24.04.1|libstemmer0d;amd64,2.2.0-4build1|libstring-
random-perl,1.10.32-2|libsvn1;amd64,1.14.3-1build1|libsystemd-shared;amd64,255.4-1ubuntu8.17|libsystemd0;amd64,255.4-1ubuntu8.17|libtalloc2;amd64,2.4.2-1build2|libtasn1-6;amd64,4.19.0-
3ubuntu0.24.04.2|libtdb1;amd64,1.4.10-1build1|libtevent0t64;amd64,0.16.1-2build1|libtiff6;amd64,4.5.1+git230720-4ubuntu2.5|libtime-date-perl,2.3300-2|libtinfo6;amd64,6.4+20240113-lubun
tu2.1|libtirpc-common,1.3.4ds-1.1build1|libtirpc3t64;amd64,1.3.4ds-1.1build1|libtcmath1;amd64,1.2.1-2build1|libtry-tiny-perl,0.131-2|libtsan2;amd64,14.2.0-4ubuntu2~24.04.1|libtsk19t
64;amd64,4.12.1+dfsg-1.1ubuntu2|libubsan1;amd64,14.2.0-4ubuntu2~24.04.1|libuchardet0;amd64,0.0.8-1build1|libudev1;amd64,255.4-1ubuntu8.17|libunistring5;amd64,1.1-2build1.1|libwind8;
amd64,1.6.2-3build1.1|liburi-perl,5.27-1|libutempter0;amd64,1.2.1-3build1|libutf8proc3;amd64,2.9.0-1build1|libuuid1;amd64,2.39.3-9ubuntu6.5|libuv1t64;amd64,1.48.0-1.1build1|libvhd11;a
md64,20210425-1build5|libvmdk1;amd64,20200926-2build5|libwbclient0;amd64,2.4.19.5+dfsg-4ubuntu2.7|libwebp7;amd64,1.3.2-0.4build3|libwhisker2-perl,2.5.1-2|libwinpr-2t64;amd64,2.11.5+df
fsg1-1build2|libwiredshark-data,4.2.2-1.1build3|libwiredshark1t64;amd64,4.2.2-1.1build3|libwiretap1t64;amd64,4.2.2-1.1build3|libwrap0;amd64,7.6.q-33|libwsutil5t64;amd64,4.2.2-1.1buil
d3|libwww-mechanize-perl,2.18-1ubuntu1|libwww-perl,6.76-1ubuntu0.1|libwww-robotrules-perl,6.02-1|libx11-6;amd64,2:1.8.7-1build1|libx11-data,2:1.8.7-1build1|libxau6;amd64,1:1.0.9-1buil
d6|libxcb-render0;amd64,1.15-1ubuntu2|libxcb-shm0;amd64,1.15-1ubuntu2|libxcb1;amd64,1.15-1ubuntu2|libxdmcp6;amd64,1:1.1.3-0ubuntu6|libxext6;amd64,2:1.3.4-1build2|libxkbfile1;amd64,1:1
.0-1build4|libxml-writer-perl,0.900-2|libxml2;amd64,2.9.14+dfsg-1.3ubuntu3.8|libxmlb2;amd64,0.3.24-1-ubuntu0.24.04.1|libxrender1;amd64,1:0.9.10-1.1build1|libxslt1.1;amd64,1:1.39-0ex
piubuntu0.24.04.3|libxtables12;amd64,1.8.10-3ubuntu2|libxshash0;amd64,0.8.2-2build1|libyaml-0.2;amd64,0.2.5-1build1|libyara10;amd64,4.5.0-1build2|libzstd1;amd64,1.5.5+dfsg2-2build1.1|
linux-libc-dev;amd64,6.8.0-137.137|lvm-16-linker-tools,1:16.0-6-2ubuntu4|login,1:4.13+dfsg1-4ubuntu3.2|logsave,1.47.0-2.4-exp1ubuntu4.1|lsb-release,12.0-2|ltrace,0.7.3-6.4ubuntu3|ly
nis,3.0.9-1-nano-db,2:12.0-4build2|naaccan,2:1.3.2+ds1-1|nawk,1.3.4.20240123-1build1|media-types,10.1.0-1aenu,2:1.50-mount,2.39.3-9ubuntu6.5|ntr-tiny,0.95-1-ubuntu0.1|mysql-common,5.8+
1.1.0build1|nano,7.2-2ubuntu0.2|nbtscan,1.7.2-2|ncat,7.94+git20230807.3be0efb1+dfsg-3build2|ncurses-base,6.4+20240113-lubuntu2.1|ncurses-bin,6.4+20240113-lubuntu2.1|net-tools,2:10.0-
linux0.4|netbase,6.4|netcat-openbsd,1.226-1ubuntu2|nikto,1:2.1.5-3.1|nmap,7.94+git20230807.3be0efb1+dfsg-3build2|nmap-common,7.94+git20230807.3be0efb1+dfsg-3build2|ntfs-3g,1:2022.
10.3-1-2ubuntu3.2|onesixtyone,0.3.4-1|openssh-client,1:9.6p1-3ubuntu1.18|openssl,3.0.13-0ubuntu2.12|packagekit,1:2.0-2ubuntu1.5|passwd,1:4.13+dfsg1-4ubuntu3.2|patchelf,0.18.0-1.1buil
d|pci-ids,0.0-2024.03.31-1ubuntu0.1|pdfcrack,0.20-1|perl,5.38.2-3-2ubuntu0.3|perl-base,5.38.2-3-2ubuntu0.3|perl-modules-5.38,5.38.2-3-2ubuntu0.3|perl-openssl-defaults;amd64,7build3|p
ngcheck,3.0.3-3|poc1-openssl-icd;amd64,5.0.2-1build3|polkitd,124-2ubuntu1.24.04.3|procps,2:4.0-4-2ubuntu2.2|publicsuffix,20231001.0357-0.1|python-apt-common,2.7.7ubuntu5.2|python3,3.12
.3-0ubuntu2.1|python3-apt,2.7.7ubuntu5.2|python3-bimalk,2.3.4+dfsg1-5|python3-blinker,1.5.0-1|python3-bss4,4.12.3-1|python3-certifi,2023.11.17-1|python3-efii,1.16.0-2build1|python3-ef
fi-backend;amd64,1.16.0-2build1|python3-charDET,5.2.0+dfsg-1|python3-charDET-normalizer,3.3.2-1|python3-click,8.1.6-2|python3-colorama,0.4.6-4|python3-cryptography,41.0.7-4ubuntu0.4|p
ython3-dbus,1.3.2-1build3|python3-defusedxml,0.7.1-2|python3-distro,1.9.0-1|python3-dnspython,2.6.1-1ubuntu1|python3-flask,3.0.2-1ubuntu1.1|python3-gi,3.48.2-1|python3-httplib2,1.20.4-
3ubuntu0.1|python3-idna,3.6-2ubuntu0.2|python3-impacket,0.11.0-2|python3-itsdangerous,2.1.2-4|python3-jinja2,3.1.2-1ubuntu1.3|python3-jwt,2.7.0-1ubuntu0.1|python3-launchpadlib,1.11.0-
6|python3-lazr-restfulclient,0.14.6-1|python3-lazr-uri,1.0.6-3|python3-lldap3,2.9.1-2|python3-lldapdomaindump,0.9.4-2|python3-ldb,2:2.8.0+samba4.19.5+dfsg-4ubuntu9.7|python3-lxml;amd64
,5.2.1-1|python3-magic,2:0.4.27-3|python3-markdown,3.5.2-1|python3-markdown-it,3.0.0-2|python3-markupsafe,2.1.5-1build2|python3-mdurl,0.1.2-1|python3-minimal,3.12.3-0ubuntu2.1|python3-
multidict,6.0.4-1.1build1|python3-netaddr,0.8.0-2ubuntu1|python3-ntlm-auth,1.5.0-1|python3-oauthlib,3.2.2-1|python3-openssl,23.2.0-1ubuntu0.1|python3-pip,24.0+dfsg1-1ubuntu1.3|python3-
pip-whl,24.0+dfsg1-1ubuntu1.3|python3-pkg-resources,68.1.2-2ubuntu2.2|python3-ply,3.11-6|python3-pyasn1,0.4.8-4ubuntu0.2|python3-pyparser,2.21-1|python3-pycryptodome,3.20.0+dfsg-1|py
thon3-pycurl,7.45.3-1build2|python3-pygments,2.17.2+dfsg-1|python3-pyparsing,3.1.1-1|python3-requests,2.31.0+dfsg1-1ubuntu1.1|python3-requests-ntlm,1.1.0-3|python3-requests-toolbelt,1.
0.0-2|python3-rich,13.7.1-1|python3-samba,2:4.19.5+dfsg1-4ubuntu9.7|python3-setuptools,68.1.2-2ubuntu2.2|python3-setuptools-whl,68.1.2-2ubuntu2.2|python3-six,1.16.0-4|python3-socks,1.7
.1+dfsg1-1|python3-software-properties,0.99.49.4|python3-soupsieve,2.5.1-1|python3-talloc;amd64,2.4.2-1build2|python3-tdb,1.4.10-1build1|python3-urllib3,2.0.7-1ubuntu0.7|python3-venv,3.1
2.3-0ubuntu2.1|python3-wadllib,1.3.6-5|python3-werkzeug,3.0.1-3ubuntu0.2|python3-wheel,0.42.0-2|python3-yaml,6.0.1-2build2|python3-yara,4.5.0-1build1|python3-zombie-imp,0.2-2|python
3.12.3,12.3.12-3-lubuntu0.15|python3.12-minimal,3.12.3-1ubuntu0.15|python3.12-venv,3.12.3-1ubuntu0.15|rake,13.0.6-3|readline-common,8.2-4build1|ripgrep,14.1.0-1|rpcsvc-proto,1.4.2-1ubuntu7
|rsync,3.2.7-1ubuntu1.5|ruby,1:3.2-2ubuntu1|ruby-addressable,2.8.5-1|ruby-ipaddress,0.8.3-3|ruby-net-telnet,0.2.0-1|ruby-public-suffix,4.0.6+ds-2|ruby-rubymaps,3.4.20-1|ruby-sdbm;amd64
,1.0.0-5build4|ruby-webrick,1.8.1-1ubuntu2|ruby-xmlrpc,0.3.2-2|ruby3.2,3.2.2-3-lubuntu0.24.04.8|rubymaps-integration,1.18|samba-common,2:4.19.5+dfsg1-4ubuntu9.7|samba-common-bin,2:4.19
.5+dfsg1-4ubuntu9.7|samba-libs;amd64,2:4.19.5+dfsg1-4ubuntu9.7|sed,4.9-2ubuntu0.24.04.1|sensible-utils,0.0.22|sgml-base,1.31|shellcheck,0.9.0-1|sleuthkit,4.12.1+dfsg1-1ubuntu2|smbclie
nt,2:4.19.5+dfsg1-4ubuntu9.7|snmp,5.9.4+dfsg1-1ubuntu3.2|socat,1.8.0-0-4ubuntu0.1|software-properties-common,0.99.49.4|sqlmap,1.8.4-1|ssllscan,2.1.2-0.1build2|steghide,0.5.1-15|strace,6
.8-0ubuntu2|sudo,1.9.15p5-3ubuntu5.24.04.2|systemd,255.4-1ubuntu8.17|systemd-dev,255.4-1ubuntu8.17|systemd-sysv,255.4-1ubuntu8.17|sysvinit-utils,3.08-6ubuntu3|tar,1.35+dfsg1-3ubuntu0.4
|tcpdump,4.99.4-3ubuntu4.24.04.1|testdisk,7.1.5-nmu1build2|tmux,3.4-1ubuntu0.1|traceroute,1:2.1.5-1|tree,2.1.1-2ubuntu3.24.04.2|tshark,4.2.2-1.1build3|tzdata,2026c-0ubuntu0.24.04.1|ub
untu-keyring,2023.11.28.1|ucf,3.0043+nmul|unminimize,0.2.1|unzip,6.0-28ubuntu1.1|util-linux,2.39.3-9ubuntu6.5|vim,2:9.1.0016-1ubuntu7.18|vim-common,2:9.1.0016-1ubuntu7.18|vim-runtime,
2:9.1.0016-1ubuntu7.18|vim-tiny,2:9.1.0016-1ubuntu7.18|wfsuzz,3.1.0-4ubuntu1|wget,1.21.4-1ubuntu4.4|whatweb,0.5.5-1|whois,5.5.22|wiredshark-common,4.2.2-1.1build3|xml-core,0.19|yara,4.5
.0-1build2|zip,3.0-13ubuntu0.2|zlib1g;amd64,1:1.3.1-3+dfsg-3-1ubuntu2.1|zsh,5.9-6ubuntu2|zsh-auto-suggestions,0.7.0-1|zsh-common,5.9-6ubuntu2|zsh-syntax-highlighting,0.7.1-2
student@student000317:~$
```

Figure 19: Lynis Report Data Output

The Lynis report data was reviewed to identify security findings and configuration weaknesses detected during the system assessment.

```
Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

=====

Exceptions found
Some exceptional events or information was found!

What to do:
You can help by providing your log file (/var/log/lynis.log).
Go to https://cisofy.com/contact/ and send your file to the e-mail address listed

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOFY - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)

student@student000317:~$ sudo cat /var/log/lynis-report.dat | grep warning
warning[ ]=PKGS-7392 Found one or more vulnerable packages.[-|-]
warning[ ]=NETW-2705 Couldn't find 2 responsive nameservers.[-|-]
warning[ ]=LOGG-2138 klogd is not running, which could lead to missing kernel messages in log files.[-|-]
student@student000317:~$
```

Figure 20: Lynis Warnings Identified During Security Assessment

The Lynis report identified security warnings, including vulnerable packages and missing kernel logging. These warnings were used to develop the security findings and recommended remediation actions below.

```
64:amd64,4.12.1-dfsg-1.1ubuntu2|libubsan1:amd64,14.2.0-4ubuntu2-24.04.1|libuchardet0:amd64,0.0.8-1build1|libudev1:amd64,255.4-1ubuntu8.17|libunistring5:amd64,1.1-2build1.1|libunwind8:amd64,1.6.2-3build1.1|liburi-perl,5.27-1|libutempter0:amd64,1.2.1-3build1|libutf8proc3:amd64,2.9.0-1build1|libuuid:amd64,2.39.3-9ubuntu5.5|libuv1t64:amd64,1.48.0-1.1build1|libvhdi1:aamd64,20210425-1build5|libvmdk1:amd64,20200926-2build5|libwbclient0:amd64,2:4.19.5-dfsg-4ubuntu9.7|libwebp7:amd64,1.3.2-0.4build3|libwhisker2-perl,2.5.1-2|libwinpr2-2t64:amd64,4.2.2-1.5-dfsg-1build2|libwiresark-data,4.2.2-1.1build3|libwiresark1t64:amd64,4.2.2-1.1build3|libwiretap1t64:amd64,4.2.2-1.1build3|libwrap0:amd64,7.6.q-33|libwsutil15t64:amd64,4.2.2-1.1build3|libwww-mechanize-perl,2.18-1ubuntu1|libwww-perl,6.76-1ubuntu0.1|libwww-robotrules-perl,6.02-1|libx11-6:amd64,2:1.8.7-1build1|libx11-data,2:1.8.7-1build1|libxau6:amd64,1:1.0.9-1build1|libxcb-render0:amd64,1.15-1ubuntu2|libxcb-shm0:amd64,1.15-1ubuntu2|libxcb1:amd64,1.15-1ubuntu2|libxdmcp6:amd64,1:1.1.3-0ubuntu6|libxext6:amd64,2:1.3.4-1build2|libxkbfile1:amd64,1:1.1.0-1build4|libxml-writer-perl,0.900-2|libxml2:amd64,2.9.14-dfsg-1.3ubuntu3.8|libxmlb2:amd64,0.3.24-1ubuntu0.24.04.1|libxrender1:amd64,1:0.9.10-1build1|libxslt1.1:amd64,1.1.39-0ex-piubuntu0.24.04.3|libxtables12:amd64,1.8.10-3ubuntu2|libxshash0:amd64,0.8.2-2build1|libyaml-0.2:amd64,0.2.5-1build1|libyara10:amd64,4.5.0-1build2|libzstd1:amd64,1.5.5+dfsg-2build1.1|linux-libc-dev:amd64,6.8.0-137.137|lvm-16-linker-tools,1:16.0.6-2ubuntu4|login,1:4.13-dfsg1-4ubuntu3.2|logsaver,1.47.0-2.4-exp1ubuntu4.1|lsb-release,12.0-2|ltrace,0.7.3-6.4ubuntu3|lynis,3.0.9-1|man-db,2.12.0-4build2|masscan,2:1.3.2+ds1-1|nawk,1:3.4.20240123-1build1|media-types,10.1.0|menu,2.1.50|mount,2.39.3-9ubuntu6.5|ntr-tiny,0.95-1.1ubuntu0.1|mysql-common,5.8+1.1.0build1|nano,7.2-2ubuntu0.2|nbtscan,1.7.2-2|ncat,7.94+git20230807.3be01efb1-dfsg-3build2|ncurses-base,6.4+20240113-1ubuntu2.1|ncurses-bin,6.4+20240113-1ubuntu2.1|net-tools,2.10-0.1ubuntu4.4|netbase,6.4|netcat-openbsd,1.226-1ubuntu2|nikto,1:2.1.5-3.1|nmap,7.94+git20230807.3be01efb1-dfsg-3build2|nmap-common,7.94+git20230807.3be01efb1-dfsg-3build2|nfs-3g,1:2022.10.3-1.2ubuntu2.3|onesixtyone,0.3.4-1|openssh-client,1:9.6p1-3ubuntu13.18|openssh,3.0.13-0ubuntu3.12|packagekit,1.2.8-2ubuntu1.5|passwd,1:4.13-dfsg1-4ubuntu3.2|patchelf,0.18.0-1build1|pci.ids,0.0-2024.03.31-1ubuntu0.1|pdfcrack,0.70-1perl,5.38.2-3ubuntu0.3|perl-base,5.38.2-3ubuntu0.3|perl-modules-5.38,5.38.2-3ubuntu0.3|perl-openssl-defaults:amd64,7build3|pngcheck,3.0.3-3|pool-openc1:amd64,5.0-2build3|poolkit,1:24-2ubuntu1.24.04.3|procp,2:4.0-4-4ubuntu2.2|publicsuffix,20231001-0357-0.1|python-apt-common,2.7.7ubuntu2.2|python3-3.12-3-0ubuntu2.1|python3-apt,2.7.7ubuntu2.2|python3-bimalk,2.3.4-dfsg1-5|python3-blinker,1.7.0-1|python3-bt4,4.12.3-1|python3-certifi,2023.11.17-1|python3-efifi,1:16.0-2build1|python3-efi-backend:amd64,1.16.0-1ubuntu1|python3-charDET,5.2.0-dfsg-1|python3-charDET-normalizer,3.3.2-1|python3-click,8.1.6-2|python3-colorama,0.4.6-4|python3-cryptography,41.0.7-4ubuntu0.4|python3-dbus,1.3.2-3build3|python3-defusedxml,0.7.1-2|python3-distro,1.9.0-1|python3-dnspython,2.6.1-1ubuntu1|python3-flask,3.0.2-1ubuntu1.1|python3-gi,3.48.2-1|python3-httplib2,0.20.4-1ubuntu0.1|python3-idna,3.6-2ubuntu0.2|python3-impacket,0.11.0-2|python3-itsdangerous,2.1.2-4|python3-jinja2,3.1.2-1ubuntu1.3|python3-jwt,2.7.0-1ubuntu2.1|python3-launchpadlib,1.11.0-6|python3-lazr-restfulclient,0.14.6-1|python3-lazr-uri,1.0.6-3|python3-lldap3,2.9.1-2|python3-lldapdomaindump,0.9.4-2|python3-ldb,2:2.8.0+samba4.19.5-dfsg-4ubuntu9.7|python3-lxml:amd64,5.2.1-1|python3-magic,2:0.4.27-3|python3-markdown,3.5.2-1|python3-markdown-it,3.0.0-2|python3-markupsafe,2.1.5-1build2|python3-mdurl,0.1.2-1|python3-minimal,3.12.3-0ubuntu2.1|python3-multidict,6.0.4-1build1|python3-netaddr,0.8.0-2ubuntu1|python3-ntlm-auth,1.5.0-1|python3-oauthlib,3.2.2-1|python3-openssl,23.2.0-1ubuntu0.1|python3-pip,24.0+dfsg-1ubuntu1.3|python3-pip-whl,24.0+dfsg-1ubuntu1.3|python3-pkg-resources,68.1.2-2ubuntu1.2|python3-ply,3.11-6|python3-pyasn1,0.4.8-4ubuntu0.2|python3-pycparser,2.21-1|python3-pycryptodome,3.20.0+dfsg-1|python3-pycurl,7.45.3-1build2|python3-pygments,2.17.2+dfsg-1|python3-pyparsing,3.1.1-1|python3-requests,2.31.0+dfsg-1ubuntu1.1|python3-requests-ntlm,1.1.0-3|python3-requests-toolbelt,1.0.0-2|python3-rich,13.7.1-1|python3-samba,2:4.19.5+dfsg-4ubuntu9.7|python3-setuptools,68.1.2-2ubuntu1.2|python3-setuptools-whl,68.1.2-2ubuntu1.2|python3-six,1.16.0-4|python3-socks,1.7.1-dfsg-1|python3-software-properties,0.99.49.4|python3-soupsieve,2.5.1-1|python3-talloc:amd64,2.4.2-1build2|python3-yaml,6.0.1-1build2|python3-yara,4.5.0-1build1|python3-zombie-imp,0.0.2-2|python3.12,3.12.3-0ubuntu0.15|python3.12-minimal,3.12.3-0ubuntu0.15|python3.12-venv,3.12.3-0ubuntu0.15|rake,13.0.6-3|readline-common,8.2-4build1|rpgrep,14.1.0-1|rpcsvc-proto,1.4.2-0ubuntu7|rsync,3.2.7-1ubuntu1.5|ruby,1:3.2-1ubuntu1|ruby-addressable,2.8.5-1|ruby-ipaddress,0.8.3-3|ruby-net-telnet,0.2.0-1|ruby-public-suffix,4.0.6+ds-2|ruby-rubygems,3.4.20-1|ruby-sdbm:amd64,1.0.0-5build4|ruby-webrick,1.8.1-1ubuntu0.2|ruby-xmlrpc,0.3.2-2|ruby3.2,3.2.3-1ubuntu0.24.04.8|rubygems-integration,1.18|samba-common,2:4.19.5+dfsg-4ubuntu9.7|samba-common-bin,2:4.19.5+dfsg-4ubuntu9.7|samba-libs:amd64,2:4.19.5+dfsg-4ubuntu9.7|sed,4.9-2ubuntu0.24.04.1|sensible-utils,0.0.22|sgml-base,1.31|shellcheck,0.9.0-1|sleuthkit,4.12.1-dfsg-1-1ubuntu2|smblclient,2:4.19.5+dfsg-4ubuntu9.7|snmp,5.9.4+dfsg-1.1ubuntu3.2|socat,1.8.0-0-4ubuntu2.1|software-properties-common,0.99.49.4|sqlmap,1.8.4-1|ssllscan,2.1.2-0-1build2|steghide,0.5.1-15|strace,6.8-0ubuntu2|sudo,1.9.15p5-3ubuntu1.24.04.2|systemd,255.4-1ubuntu0.17|systemd-dev,255.4-1ubuntu0.17|systemd-sysv,255.4-1ubuntu0.17|sysvinit-utils,3.08-6ubuntu3|tar,1.35+dfsg-3ubuntu0.4|tcpdump,4.99.4-3ubuntu0.24.04.1|testdisk,7.1.5-nmu1build2|tmux,3.4-1ubuntu0.1|traceroute,1:2.1.5-1|tree,2.1.1-2ubuntu0.24.04.2|tshark,4.2.2-1build3|tzdata,2023c-0ubuntu0.24.04.1|ubuntukyring,2023.11.28-1|ucf,3.0043+nmul|unminimize,0.2.1|unzip,6.0-28ubuntu4.1|util-linux,2.39.3-9ubuntu5.5|vim,2:9.1.0016-1ubuntu7.18|vim-common,2:9.1.0016-1ubuntu7.18|vim-runtime,2:9.1.0016-1ubuntu7.18|vim-tiny,2:9.1.0016-1ubuntu7.18|wifuzz,3.1.0-1ubuntu1|wget,1.21.4-1ubuntu4.4|whatweb,0.5.5-1|whois,5.5.22|wiresark-common,4.2.2-1.1build3|xmll-core,0.19|yara,4.5.0-1build2|zip,3.0-13ubuntu2.2|zlib:amd64,1:1.3-dfsg-3.1ubuntu2.1|zsh,5.9-6ubuntu2|zsh-auto suggestions,0.7.0-1|zsh-common,5.9-6ubuntu2|zsh-syntax-highlighting,0.7.1-2

student@student000317:~$ sudo grep -i warning /var/log/lynis.log
2026-10-01 20:57:33 Warning: Found one or more vulnerable packages. [test:PKGS-7392] [details:-] [solution:-]
2026-10-01 20:57:39 Warning: Couldn't find 2 responsive nameservers [test:NETW-2705] [details:-] [solution:-]
2026-10-01 20:57:41 Warning: klogd is not running, which could lead to missing kernel messages in log files [test:LOGG-2138] [details:-] [solution:-]
2026-10-01 20:57:42 Result: Skipping display warning, as virtual machines usually don't need time synchronization in the VM itself
2026-10-01 20:57:47 Skipped test CONT-8104 (Checking Docker info for any warnings)
student@student000317:~$
```

Figure 21: Lynis Security Warnings

The Lynis log identified several security warnings, including vulnerable packages, DNS name server issues, and missing kernel logging. These warnings were reviewed to determine their security impact and required remediation.

Finding F03 – Vulnerable Packages Identified

Evidence

Lynis reported:

Found one or more vulnerable packages

Risk Rating

High

Control Owner

Infrastructure Administrator

Recommendation

Apply updates and establish a regular patch-management process to reduce exposure to known vulnerabilities.

Finding F04 – Kernel Logging Weakness

Evidence

Lynis reported:

klogd is not running, which could lead to missing kernel messages in log files

Risk Rating

High

Control Owner

Linux Administrator

Recommendation

Enable kernel logging services to improve visibility into system-level events.

```
2026-10-01 20:57:21 Suggestion: Install needrestart, alternatively to debian-goodies, so that you can run needrestart after upgrades to determine which daemons are using old versions of libraries and need restarting. [test:DEB-0831] [details:-] [solution:-]
2026-10-01 20:57:21 Suggestion: Install fail2ban to automatically ban hosts that commit multiple authentication errors. [test:DEB-0880] [details:-] [solution:-]
2026-10-01 20:57:21 Suggestion: Determine runlevel and services at startup [test:BOOT-5180] [details:-] [solution:-]
2026-10-01 20:57:22 Suggestion: Consider hardening system services [test:BOOT-5264] [details:Run '/usr/bin/systemd-analyze security SERVICE' for each service] [solution:-]
2026-10-01 20:57:22 Suggestion: Determine why /vmlinuz or /boot/vmlinuz is missing on this Debian/Ubuntu system. [test:KRN1-5788] [details:/vmlinuz or /boot/vmlinuz] [solution:-]
2026-10-01 20:57:22 Suggestion: If not required, consider explicit disabling of core dump in /etc/security/limits.conf file [test:KRN1-5820] [details:-] [solution:-]
2026-10-01 20:57:22 Suggestion: Check the output of ps for dead or zombie processes [test:PROC-3612] [details:-] [solution:-]
2026-10-01 20:57:22 Suggestion: Configure password hashing rounds in /etc/login.defs [test:AUTH-9230] [details:-] [solution:-]
2026-10-01 20:57:22 Suggestion: Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc [test:AUTH-9262] [details:-] [solution:-]
2026-10-01 20:57:22 Suggestion: Look at the locked accounts and consider removing them [test:AUTH-9284] [details:-] [solution:-]
2026-10-01 20:57:22 Suggestion: Configure minimum password age in /etc/login.defs [test:AUTH-9286] [details:-] [solution:-]
2026-10-01 20:57:22 Suggestion: Configure maximum password age in /etc/login.defs [test:AUTH-9286] [details:-] [solution:-]
2026-10-01 20:57:23 Suggestion: Default umask in /etc/login.defs could be more strict like 027 [test:AUTH-9328] [details:-] [solution:-]
2026-10-01 20:57:23 Suggestion: To decrease the impact of a full /home file system, place /home on a separate partition [test:FILE-6310] [details:-] [solution:-]
2026-10-01 20:57:23 Suggestion: To decrease the impact of a full /var file system, place /var on a separate partition [test:FILE-6310] [details:-] [solution:-]
2026-10-01 20:57:24 Suggestion: Disable drivers like USB storage when not used, to prevent unauthorized storage or data theft [test:USB-1000] [details:-] [solution:-]
2026-10-01 20:57:24 Suggestion: Disable drivers like firewall storage when not used, to prevent unauthorized storage or data theft [test:STRG-1846] [details:-] [solution:-]
2026-10-01 20:57:24 Suggestion: Check DNS configuration for the dns domain name [test:NAME-4028] [details:-] [solution:-]
2026-10-01 20:57:26 Found package: zsh-autosuggestions (version: 0.7.0-1)
2026-10-01 20:57:26 Suggestion: Install debsums utility for the verification of packages with known good database. [test:PKGS-7370] [details:-] [solution:-]
2026-10-01 20:57:33 Suggestion: Update your system with apt-get update, apt-get upgrade, apt-get dist-upgrade and/or unattended-upgrades [test:PKGS-7392] [details:-] [solution:-]
2026-10-01 20:57:33 Suggestion: Install package apt-show-versions for patch management purposes [test:PKGS-7394] [details:-] [solution:-]
2026-10-01 20:57:38 Suggestion: Consider using a tool to automatically apply upgrades [test:PKGS-7420] [details:-] [solution:-]
2026-10-01 20:57:39 Suggestion: Check your resolv.conf file and fill in a backup nameserver if possible [test:NETW-2705] [details:-] [solution:-]
2026-10-01 20:57:39 Suggestion: Determine if protocol 'dccp' is really needed on this system [test:NETW-3200] [details:-] [solution:-]
2026-10-01 20:57:39 Suggestion: Determine if protocol 'sctp' is really needed on this system [test:NETW-3200] [details:-] [solution:-]
2026-10-01 20:57:39 Suggestion: Determine if protocol 'rds' is really needed on this system [test:NETW-3200] [details:-] [solution:-]
2026-10-01 20:57:39 Suggestion: Determine if protocol 'tipc' is really needed on this system [test:NETW-3200] [details:-] [solution:-]
2026-10-01 20:57:41 Suggestion: Check if any syslog daemon is running and correctly configured. [test:LOGG-2130] [details:-] [solution:-]
2026-10-01 20:57:41 Suggestion: Add a legal banner to /etc/issue, to warn unauthorized users [test:BANN-7126] [details:-] [solution:-]
2026-10-01 20:57:41 Suggestion: Add legal banner to /etc/issue.net, to warn unauthorized users [test:BANN-7130] [details:-] [solution:-]
2026-10-01 20:57:42 Suggestion: Enable process accounting [test:ACCT-9622] [details:-] [solution:-]
2026-10-01 20:57:42 Suggestion: Enable sysstat to collect accounting (no results) [test:ACCT-9626] [details:-] [solution:-]
2026-10-01 20:57:42 Suggestion: Enable auditd to collect audit information [test:ACCT-9628] [details:-] [solution:-]
2026-10-01 20:57:47 Suggestion: Install a file integrity tool to monitor changes to critical and sensitive files [test:FWNT-4350] [details:-] [solution:-]
2026-10-01 20:57:47 Suggestion: Determine if automation tools are present for system management [test:TOOL-5002] [details:-] [solution:-]
2026-10-01 20:57:48 Suggestion: Consider restricting file permissions [test:FILE-7524] [details:See screen output or log file] [solution:text:Use chmod to change file permissions]
2026-10-01 20:57:49 Suggestion: One or more sysctl values differ from the scan profile and could be tweaked [test:KRN1-6000] [details:] [solution:Change sysctl value or disable test ( skip test:KRN1-6000:(sysctl keys))]
2026-10-01 20:57:49 Suggestion: Harden compilers like restricting access to root user only [test:HRDN-7222] [details:-] [solution:-]
2026-10-01 20:57:49 Suggestion: Harden the system by installing at least one malware scanner, to perform periodic file system scans [test:HRDN-7230] [details:-] [solution:Install a tool like rkhunter, chkrootkit, OSSEC]
student@student000317:~$
```

Figure 22: Lynis Security Hardening Suggestions

Lynis generated several hardening suggestions to improve the system's security posture. Relevant recommendations included strengthening password controls, implementing file integrity monitoring, improving audit logging, applying system updates, and deploying malware scanning.

Finding F05 – Weak Password Controls

Evidence

Lynis recommended:

Configure minimum password age
Configure maximum password age
Install password strength testing

Risk Rating

High

Control Owner

Identity Administrator

Recommendation

Implement password-aging controls and password-strength validation to improve authentication security.

Finding F06 – Missing File Integrity Monitoring

Evidence

Lynis recommended:

Install a file integrity tool to monitor changes to critical and sensitive files

Risk Rating

High

Control Owner

Security Administrator

Recommendation

Deploy a file-integrity monitoring solution such as AIDE or OSSEC.

Finding F07 – Malware Protection Not Implemented

Evidence

Lynis reported:

Malware Scanner [X]

Risk Rating

Medium

Control Owner

Security Administrator

Recommendation

Implement a malware-scanning solution to strengthen endpoint protection.

6. Control Monitoring and Governance Mapping

Control Monitoring Table

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
Audit Event Monitoring	Auditd installed, audit rules configured, but audit.log was unavailable and ausearch/aureport could not retrieve records	Linux Administrator	Needs Improvement	Audit logs should be accessible and generated continuously	High. Security events cannot be fully reviewed or investigated without audit records	Verify Auditd configuration and ensure audit logs are generated and retained	Run ausearch and aureport again and confirm records are returned
Authentication Monitoring	auth.log and syslog were not available in the environment	Linux Administrator	Needs Improvement	Authentication events should be available for review	High. Reduced visibility into login activity and privileged access events	Enable and retain authentication logging	Verify auth.log availability and confirm authentication events are recorded
Vulnerability Management	Lynis warning: "Found one or more vulnerable packages"	Infrastructure Administrator	Deficient	Zero critical vulnerabilities and timely patching of systems	High. Vulnerable software may be exploited if not updated	Apply operating system and package updates	Run apt update/upgrade and rerun Lynis to verify warning is cleared
Password Management	Lynis recommended minimum password age, maximum password age, and password-strength testing	Identity Administrator	Needs Improvement	Password policy should enforce aging and strength requirements	High. Weak password controls increase the risk of unauthorized access	Configure password-aging settings and password-strength controls	Verify configuration in login.defs and review Lynis results
File Integrity Monitoring	Lynis recommended installation of a file-integrity monitoring tool	Security Administrator	Deficient	Critical file changes should be monitored and reported	High. Unauthorized modifications may go undetected	Implement AIDE, OSSEC, or another file-integrity solution	Verify monitoring reports and confirm file-change alerts are generated

Control / Objective	Evidence	Owner	Observed Status	KPI/KRI or Threshold	Risk / Significance	Remediation	Retest / Follow-up
Malware Protection	Lynis result showed "Malware Scanner [X]"	Security Administrator	Needs Improvement	Malware protection should be installed and active	Medium. Malware infections may not be detected promptly	Install and maintain a malware-scanning solution	Rerun Lynis and confirm malware scanner is detected
Kernel Logging	Lynis warning: "klogd is not running"	Linux Administrator	Needs Improvement	Kernel events should be logged and available for review	High. Important kernel-level events may be missed	Enable kernel logging services	Review kernel logs and verify logging functionality
Secure Configuration Management	Lynis identified multiple hardening recommendations including file permissions, legal banner, and system-hardening settings	Linux Administrator	Needs Improvement	Continuous improvement of system-hardening score	Medium. Weak configurations increase the attack surface	Implement recommended system-hardening measures	Rerun Lynis and compare hardening score with baseline (62)

7. Governance Escalation Analysis

Highest-Priority Finding

The highest-priority finding is the presence of vulnerable packages identified during the Lynis assessment. Vulnerable software presents an immediate risk because known weaknesses may be exploited if security updates are not applied.

Ownership of Findings

Findings	Owner
Missing audit logs	Linux Administrator
Missing authentication logs	Linux Administrator
Vulnerable packages	Infrastructure Administrator
Weak password controls	Identity Administrator
Missing file integrity monitoring	Security Administrator
Malware protection gap	Security Administrator

Escalation Thresholds

Escalation should occur when:

- Audit logs remain unavailable for more than 24 hours.
- Authentication logs are unavailable.
- High-risk vulnerabilities remain unresolved.
- Critical controls fail retesting.
- Security-hardening recommendations remain unaddressed.

Evidence of Successful Remediation

Successful remediation would be demonstrated through:

- Available audit and authentication logs
- Successful ausearch and aureport results
- Improved Lynis hardening score
- Removal of vulnerable-package warnings
- Presence of file-integrity monitoring
- Presence of malware-scanning capability

8. SIEM and Continuous Monitoring Mapping

Data Source	Monitoring Purpose	SIEM Collection	Escalation Priority
Auditd	Audit-event monitoring	Yes	High
ausearch	Audit investigations	Yes	High
aureport	Audit reporting	Yes	Medium
journalctl	System monitoring	Yes Yes	Medium
Authentication Logs	Login monitoring	Yes	High
Lynis Reports	Security posture monitoring	Yes	Medium

9. Remediation and Retest Plan

Findings	Recommended Action	Owner	Retest Method
Vulnerable Packages	Apply security updates	Infrastructure Administrator	Re-run Lynis
Missing Audit Logs	Restore audit logging	Linux Administrator	Verify ausearch output
Missing Authentication Logs	Restore authentication logging	Linux Administrator	Review available logs
Weak Password Controls	Configure password policies	Identity Administrator	Verify configuration settings
Missing File Integrity Monitoring	Install AIDE or OSSEC	Security Administrator	Verify monitoring reports
Malware Protection Gap	Install malware scanner	Security Administrator	Verify Lynis detection
Kernel Logging Weakness	Enable kernel logging	Linux Administrator	Review kernel log output

10. Conclusion

The assessment confirmed that monitoring and auditing tools were installed and available within the Ubuntu environment. Custom audit rules were successfully configured and test activities were performed to generate security events. However, several validation activities were affected by limitations within the training environment.

Where audit records or log files were unavailable, observations were recorded as environmental constraints rather than automatically being classified as control failures. Despite these limitations, the assessment successfully identified several areas requiring improvement.

The Lynis security assessment highlighted vulnerable packages, weak password controls, missing file-integrity monitoring, insufficient kernel logging, and the absence of malware protection. The overall Hardening Index score of 62 indicates moderate security maturity and demonstrates the need for additional hardening measures.

Implementation of the recommended actions, followed by formal retesting, will improve security monitoring, governance effectiveness, and the overall security posture of the environment.

Appendix A – Screenshot Evidence Register

Figure	Evidence Description
Figure 1	Auditd Installation Verification
Figure 2	Auditctl Binary Verification
Figure 3	Root Access Validation
Figure 4	Custom Audit Rule Configuration
Figure 5	Auditctl Operation Not Permitted
Figure 6	Security Event Generation Activity
Figure 7	Ausearch Verification
Figure 8	Aureport Verification
Figure 9	Missing audit.log Evidence
Figure 10	journalctl No Entries
Figure 11	/var/log Directory Review
Figure 12	Empty Audit Directory
Figure 13	Missing auth.log
Figure 14	Missing syslog
Figure 15	Empty Journal Directory
Figure 16	Last Command Output
Figure 17	Lynis Installation
Figure 18	Lynis Version Verification
Figure 19	Lynis Final Results (Hardening Index 62)
Figure 20	Lynis Report Data
Figure 21	Vulnerable Package Warning
Figure 22	Additional Lynis Warnings
Figure 23	Lynis Security Recommendations